

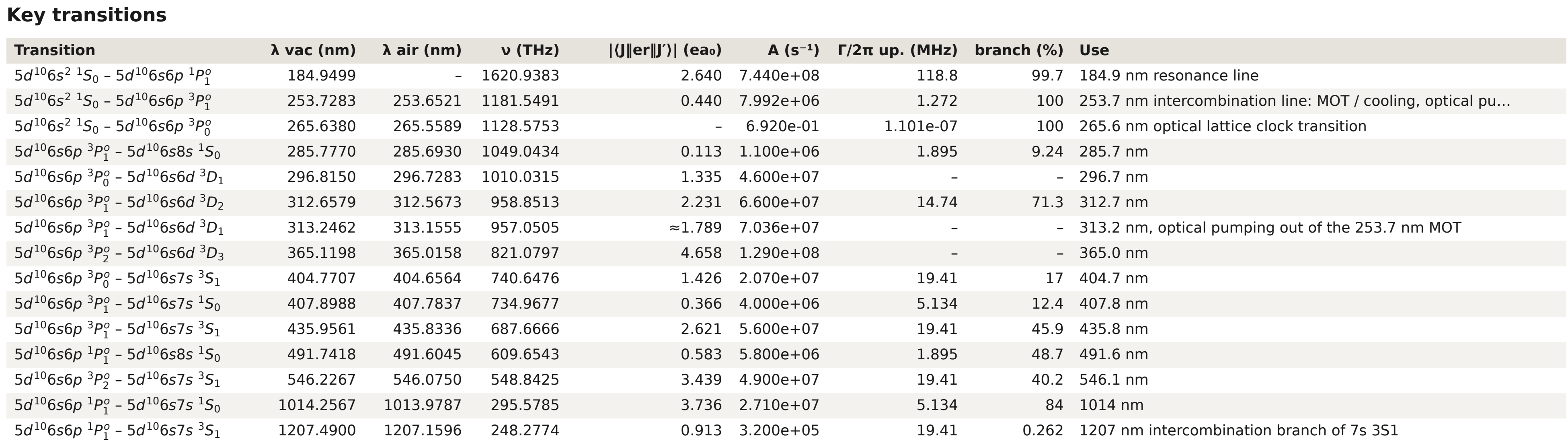
¹⁹⁹Hg (Hg I), nuclear spin $I = 1/2$. Strongest classified transitions below 2 μm (NIST): 27 levels, 47 lines.

Hg⁺ ionisation limit 84184.15 cm⁻¹ = 10.43750 eV (118.787 nm)

Arrow colour = vacuum wavelength (nm)



300 400 500 600 700 800 1000 1200 1500 2000



λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$	253.728	196-198	4179.95(3)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$	253.728	200-198	-4805.52(4)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_1^o$	253.728	202-198	-10101.224(2)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_2^o$	253.728	204-198	-15311.60(4)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_0^o$	253.728	199-198	-652(6)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_0^o$	253.728	201-198	-6410(6)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_0^o$	265.638	199-201	5729.668(8)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P_0^o$	265.638	198-199	654.48
$5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_2$	312.658	196-198	-289.4(12)
$5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_2$	312.658	200-198	328.0(6)
$5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_2$	312.658	202-198	686.7(6)
$5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_1$	312.658	204-198	1038.3(6)
$5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_1$	313.246	196-198	-307.5(10)
$5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_1$	313.246	200-198	347.2(6)
$5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_1$	313.246	202-198	725.5(6)
$5d^{10}6s6p\ ^3P_1^o - 5d^{10}6s6d\ ^3D_1$	313.246	204-198	1097.8(6)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$